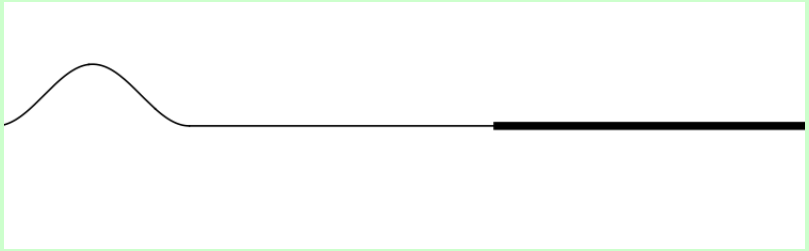


Standing Waves

Last time: pulse on string

$$A_R = \frac{Z_I - Z_2}{Z_1 + Z_2} A_I$$

$$A_T = \frac{2Z_I}{Z_1 + Z_2} A_I$$



If medium 2 denser, inverted reflection



If medium 2 less dens, no inversion

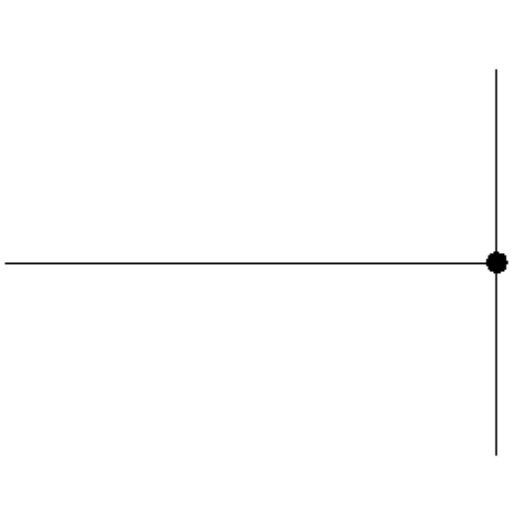
$$A_R = \frac{Z_I - Z_2}{Z_1 + Z_2} A_I$$

$$A_T = \frac{2Z_I}{Z_1 + Z_2} A_I$$

Two extreme solutions: fixed end vs. free end

Fixed End

Like an inf. thick string



$$Z_2 \to \infty$$

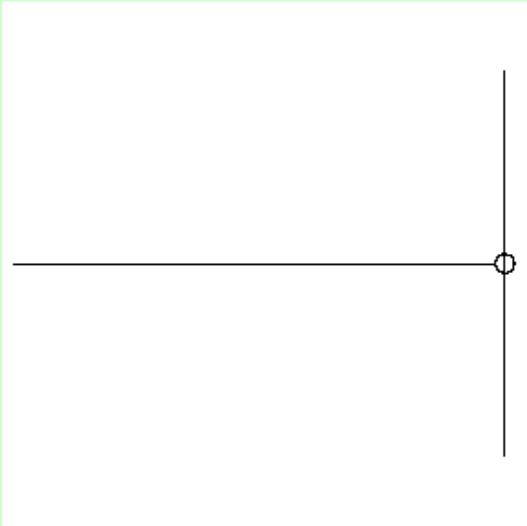
(L'hopital)

$$A_R = \frac{Z_1 - \infty}{Z_1 + \infty} A_i = \frac{-\infty}{\infty} A_I = \boxed{-A_I}$$

$$A_T = \frac{2Z_1}{Z_1 + \infty} A_i = \frac{2}{1 + \infty/Z_1} A_I = \boxed{0}$$

Free End

Like an inf. thin string



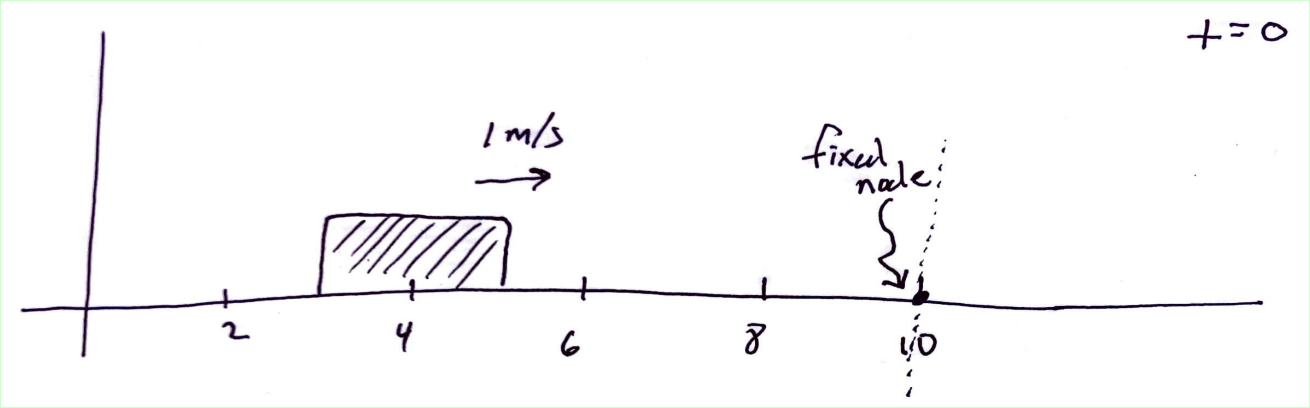
Similarly, you will find :

$$A_R = A_i$$

$$A_T = 2A_i$$

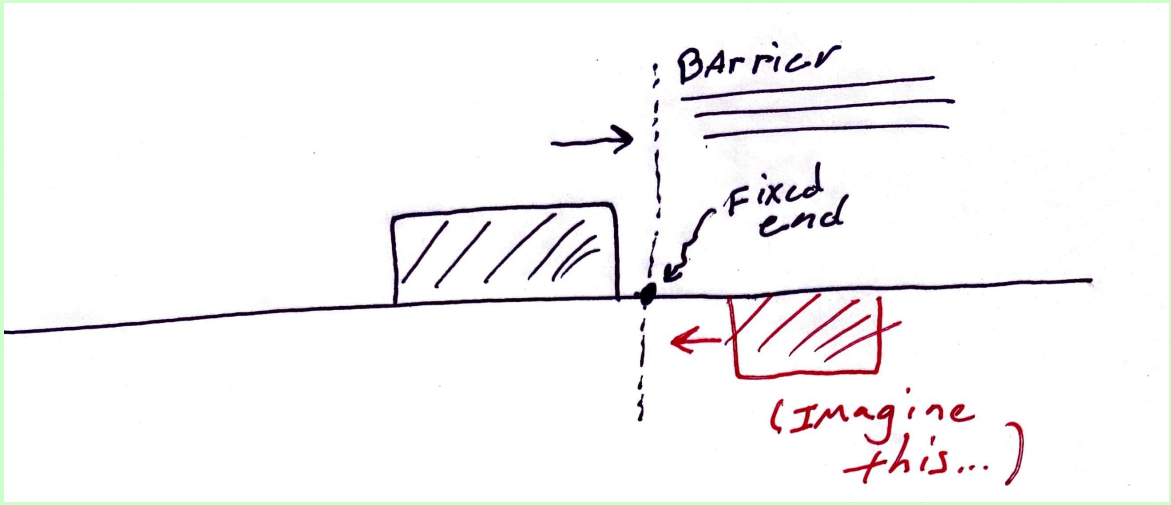
Demo: Slinky

Ex. A square wave pulse is travelling to the right at 1 m/s. At x = 10 m, it encounters a fixed end (or node). **Draw snapshots of the wave at t = 5, 6, and 7 seconds.**

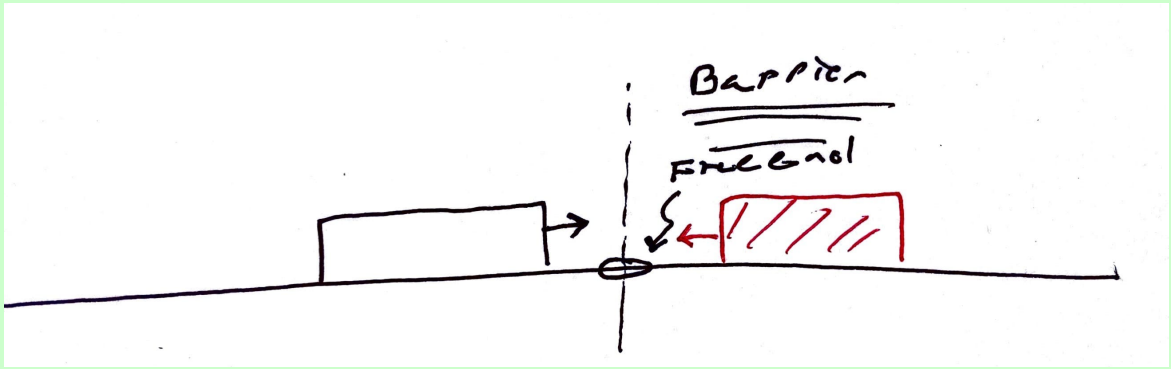


B. Repeat the exercise but now for a free end (anti-node) at x = 10

For **Fixed end** reflection, use superposition of the wave with its inversion going to –x

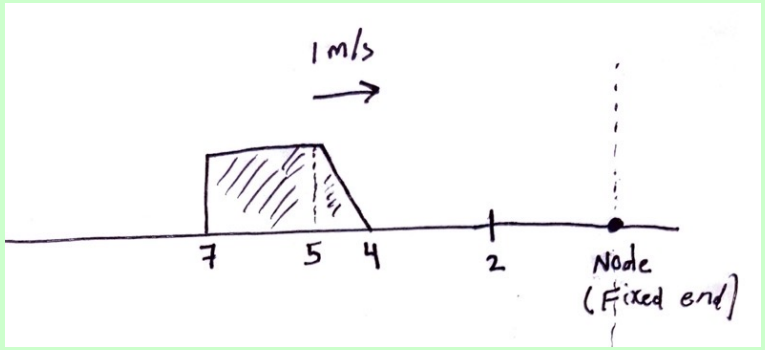


For **Free end** reflection, use superposition of the wave with its mirror going to –x



Ex. An irregularly shaped wave-pulse is incident from the right as shown.

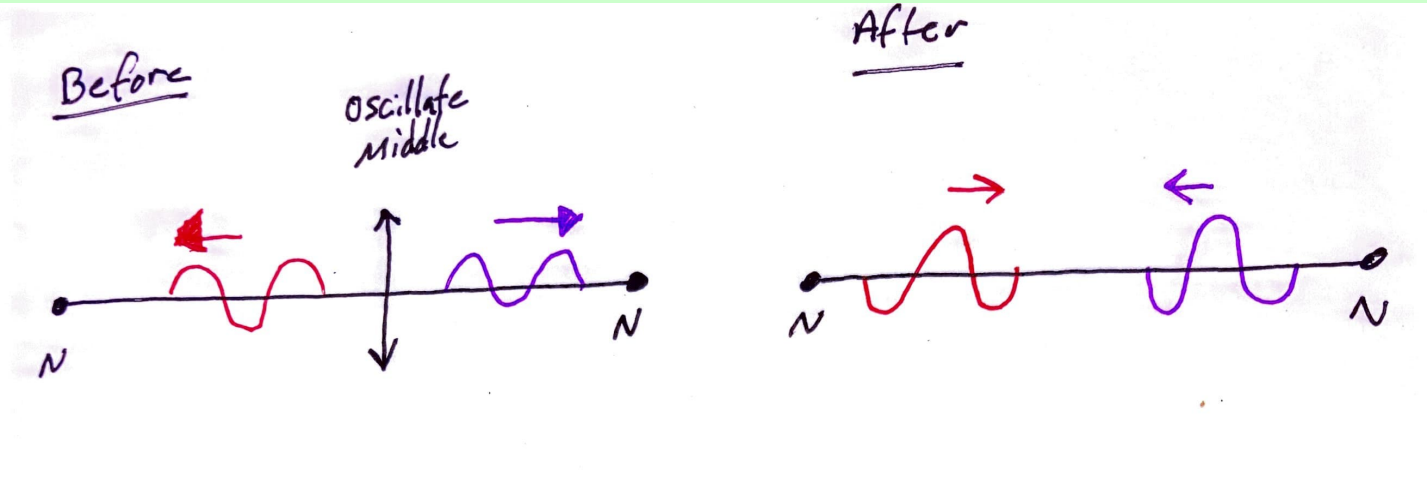
t = 0 s



A. Extend the drawing to include the incoming imaginary reflected pulse at t=0

B. Draw snapshot graph of the overall wave at t=4, 5, and 8 seconds.

Standing Waves: If there are two nodes, a standing wave can be formed. Start by oscillating the middle. The two waves will travel to the nodes, reflect, and then superpose back in the middle.



Right traveling wave $D_R = a\sin(kx - \omega t)$ $D_L = a\sin(kx + \omega t)$ **Left traveling wave**

$$D_{\text{total}} = \sum D_i = a\sin(kx - \omega t) + a\sin(kx + \omega t)$$

Use Trig Identity: $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

$$D_{\text{total}} = (a\sin kx\cos\omega t - a\cos kx\sin\omega t) + (a\sin kx\cos\omega t + a\cos kx\sin\omega t)$$

$$D_{\text{total}} = \boxed{2a\sin(kx)\cos(\omega t)}$$ Standing wave equation

$$D_{\text{total}} = \boxed{2a\sin(kx)\cos(\omega t)}$$

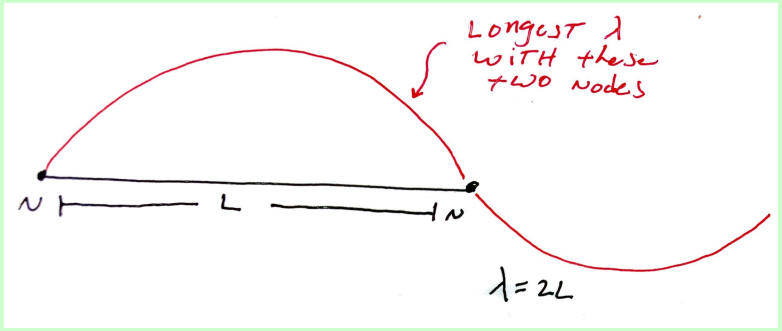
People usually break this up…

$$A(x) = 2a\sin(kx) \qquad D_{\text{total}} = A(x)\cos(\omega t)$$

Amplitude function

The A(x) shows you what the standing wave looks like at a particular moment in time. To compute it, you need to know which k you are talking about…

The longest possible wavelength would have $\lambda = 2L$



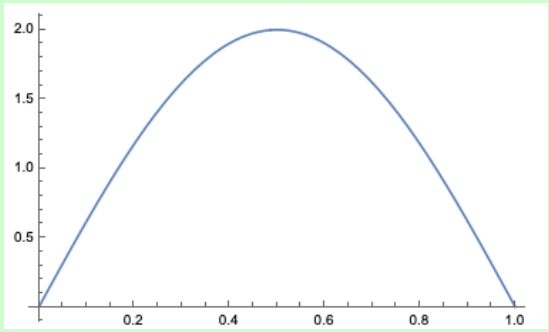
$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{2L} = \frac{\pi}{L}$$

$$A(x) = 2a\sin(kx) = \boxed{2a\sin\left(\frac{\pi}{L}x\right)}$$

Ex. The longest possible standing wave is excited on a 1 m stretch of wire. The standing wave was made by oscillating a string such that a right and left moving traveling wave were created, each with an amplitude of 1 m.

A. Use mathematica to plot the Amplitude function at t =0 seconds.

$$A(x) = 2a\sin(kx) = 2a\sin\left(\frac{\pi}{L}x\right) = 2\sin(\pi x)$$



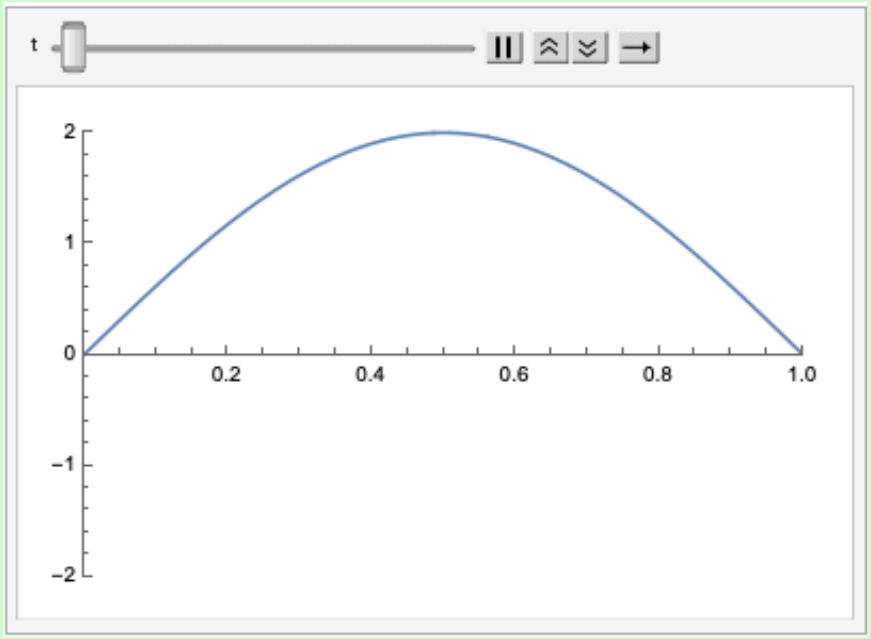
The equation of a standing wave $D_{\text{total}} = A(x)\cos(\omega t)$

The picture above is for t = 0. It's the initial configuration of the standing wave.

Recall that: $\omega = 2\pi f = \dfrac{2\pi}{P}$ So,

$$\boxed{D_{\text{total}} = A(x)\cos\left(\frac{2\pi}{P}t\right)}$$

B. Using the amplitude function from the previous page, then animating over a full period.

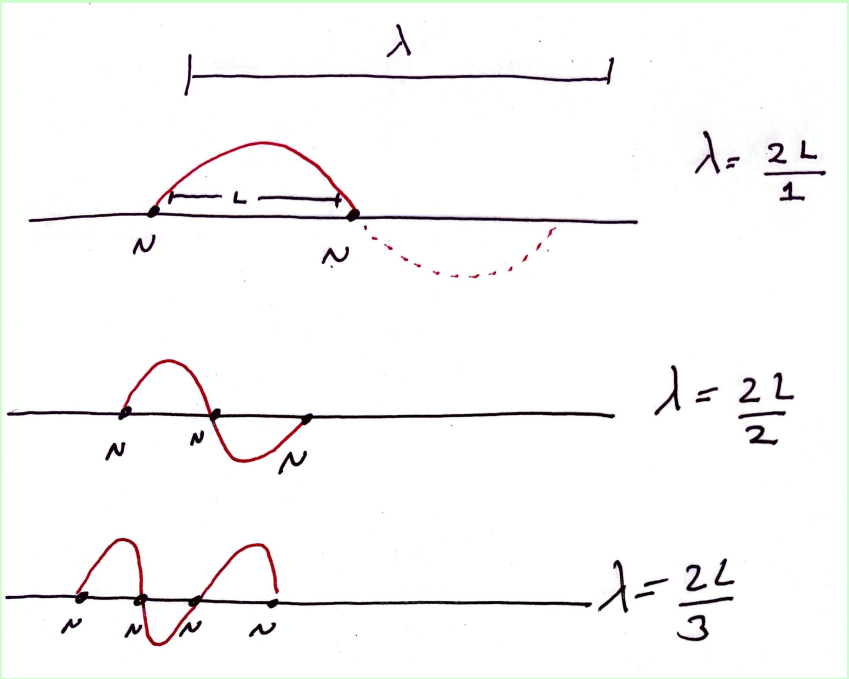


To recap, $$D_{\text{total}} = \boxed{2a\sin(kx)\cos(\omega t)}$$

Everything before the cos(wt) is the amplitude function.
Figure out which k fits between your nodes. This is your initial amplitude function
Then run it forward in times with cos(wt). You will have to rewrite omega in terms of p or f

Which wavelengths fit between nodes?

$$\lambda_n = \frac{2L}{n}$$



Which frequencies fit between nodes?

$$v = \lambda_n f = \left(\frac{2L}{n}\right) f$$

$$f = n\left(\frac{V}{2L}\right) = \boxed{nf_1}$$

f_1 is the fundamental frequency

Lets say its 10 Hz, then you will
See standing waves at every
Integer multiple of 10: eg. 10,20,30 hz

Wave demo

Ex. A 2 m length of wire is clamped down on either end. The mass density of the string is 0.1 kg/m, and the string has a tension of 100 N applied to it.

A. What is the wavelength of the longest standing wave?

B. What is the wave velocity on the string?

C. Which frequencies should you oscillate the string with?

D. Write out the full standing wave equation